# Flux Workflow

## Automation Platform

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**Flux Workflow** is a self-hosted, enterprise-grade platform for visual, no-code workflow automation. It lets teams design, run, and monitor multi-step processes, eliminating manual work across repetitive, cross-system tasks. Built on a secure internal infrastructure, it connects existing tools (like Gmail, Slack, and Basecamp) into intelligent, AI-powered pipelines. Workflows trigger automatically by schedule, event, or external signal, and every execution is fully logged and auditable. By standardizing and accelerating operations, Flux Workflow replaces manual effort with consistency and reliability, saving productive team hours weekly with zero recurring vendor cost.

# Overview / Identifying the Problem

### 1. What is the manual process today?

Across the IT team and connected departments, repetitive operational tasks are being performed entirely by hand. This includes:

- Manually reading Gmail inboxes, deciding who to notify, and forwarding or acting on messages one by one.
- Copying updates from one tool (e.g., a job completion email) and manually posting status messages to Slack, Microsoft Teams, or Basecamp.
- Manually pulling data from Google Sheets or Docs, aggregating it, and distributing summaries to stakeholders.
- Manually querying external systems via HTTP endpoints and transcribing results into reports or tickets.
- Periodically checking systems or inboxes on a fixed cadence with no automated trigger, relying on human memory and discipline.
- Manually prompting ChatGPT or similar tools with raw data to get summaries or decisions, then manually acting on those results.

## 2. How long does it take, and what does the ideal outcome look like?

**Ideal outcome:** Any multi-step, repetitive process that currently requires a person to manually connect systems, move data, or send notifications is replaced by a visual, testable, auditable workflow that runs automatically or triggered by a schedule, a webhook, or a real-time app event.

| Task | Current Manual Time | Ideal Automated Time |
|---|---|---|
| Email triage + notification | 20–40 min/day per person | < 1 min (triggered automatically) |
| Cross-platform status update | 10–15 min per event | Instant, no human touch |
| Data report aggregation | 30–60 min per report cycle | Runs on schedule, delivered automatically |
| External API lookup + action | 15–30 min per lookup | Seconds, no human involvement |
| AI summarization + routing | 15–20 min per task | Automatic, with LLM node inline |

# Proposed Solution

## 1. Tool and Approach

The solution is **Flux Workflow**, a self-hosted visual workflow automation platform built internally. It is modeled as a private alternative to tools like n8n or Zapier with full infrastructure control and no vendor lock-in.

**Backend**

- Runtime: Node.js 20+, TypeScript
- HTTP framework: Fastify 5
- Database: MongoDB 7 (workflows, executions, credentials, users)
- Job queue: Redis 7 + BullMQ (reliable async execution)
- Scheduling: node-cron (cron-based triggers)
- Validation: Zod
- Auth: JWT + Google OAuth (@fastify/jwt, google-auth-library)
- AI providers: OpenAI, Anthropic Claude, Google Gemini, Meta Llama (via @anthropic-ai/sdk, openai, @google/generative-ai)

**Frontend**

- Framework: React 18 + TypeScript, built with Vite 6
- Canvas / Visual Builder: @xyflow/react (React Flow)
- State management: Zustand
- Data fetching: TanStack Query
- Styling: Tailwind CSS

**Deployment**: Docker-based (single image serving the full stack), with docker-compose bundling the app, MongoDB, and Redis. Fully self-hosted on internal infrastructure.

## 2. What specific tasks or workflows will it affect?

- Automated email monitoring and routing (Gmail triggers)
- Scheduled data aggregation from Google Sheets/Docs/Drive
- AI-powered summarization and classification of incoming data
- Real-time notifications to Slack, Microsoft Teams, and Basecamp
- HTTP-based integrations with any internal or external API
- Webhook-driven automations triggered by third-party systems
- Conditional branching logic (if/else, switch) replacing manual decision-making

## 3. Who on the team or other departments will use it?

- IT team for the Internal tools monitoring, primary builders and owners of workflows
- Any department that currently sends manual status updates on Gmail, Slack, Teams, or Basecamp users who receive automated notifications
- Stakeholders who receive reports. Instead of waiting for a human to compile and send, reports arrive on schedule

**Access is controlled:** new users are pending until an owner approves them, ensuring only authorized personnel can build or trigger workflows.

## 4. What data sources will the tool access?

- MongoDB for internal platform data (workflow definitions, execution history, credentials, user accounts)
- Redis for job queue state (BullMQ)
- Google Workspace APIs for Gmail (read/send), Google Drive (files), Google Docs, Google Sheets
- Slack API that send messages, read events via polling
- Microsoft Graph API for Microsoft Teams messages and events
- Basecamp 3 API for project updates, messages via OAuth + polling
- Any HTTP endpoint for HTTP nodes allow calling any REST API (internal or external)
- LLM APIs for OpenAI, Anthropic, Google Gemini, Meta Llama (for AI-powered steps within workflows)

Credentials for all integrations are stored securely in MongoDB and referenced by workflows without exposing raw secrets in the UI.

## 5. How will the accuracy of results be verified?

- Each node can be tested individually before deploying a full workflow
- Every workflow run is stored with full step-by-step output in MongoDB, visible in the execution log panel
- Inbound webhook triggers are verified with a shared secret to prevent spoofed payloads
- All API inputs and workflow node configurations are validated at the boundary
- The UI includes a version history modal, allowing rollback or comparison of workflow changes
- Manual triggers allow running a workflow on demand to confirm output before scheduling it

6. **How much in resource, usage, or subscription costs does this require to run?**

Flux Workflow is self-hosted and MIT-licensed, meaning there is no platform license fee or per-seat subscription. Costs fall into two categories:

**Infrastructure (Fixed, One-Time Setup)**

The platform runs entirely via Docker and requires a server (Railway) capable of running three containers simultaneously:

| Component | Requirement |
|---|---|
| App container | Node.js 20+, ~256–512 MB RAM |
| MongoDB 7 | ~512 MB RAM + persistent disk storage |
| Redis 7 (Alpine) | ~50–100 MB RAM |

A modest server (2 vCPU, 2 GB RAM) is sufficient for small-to-medium team usage.

**External API Costs (Variable, Pay-Per-Use - Only When Used)**

These costs only apply if the corresponding node type is used in a workflow:

| Service | Cost Model |
|---|---|
| OpenAI (GPT-4o, etc.) | Per token - ~$0.01–$0.06 per 1K tokens depending on model |
| Anthropic Claude | Per token - ~$0.003–$0.015 per 1K tokens |
| Google Gemini | Free tier available; pay-per-use beyond quota |
| Gmail / Drive / Docs / Sheets | No additional cost - included in your existing Google Workspace subscription |

| | |
|---|---|
| Microsoft Teams (Graph API) | No additional cost — included in your existing Microsoft 365 subscription |
| Slack API | No additional cost — included within standard Slack plan rate limits |
| Basecamp API | No additional cost — included in your existing Basecamp subscription |

**Practical bottom line:** The only new recurring cost is LLM API usage (OpenAI/Anthropic/Gemini), and only if AI nodes are used in workflows. All other integrations are already covered by existing subscriptions. For workflows that do not use AI, the marginal cost is effectively zero.

## 7. Does this tool and the AI get full access to all items in Google Drive?

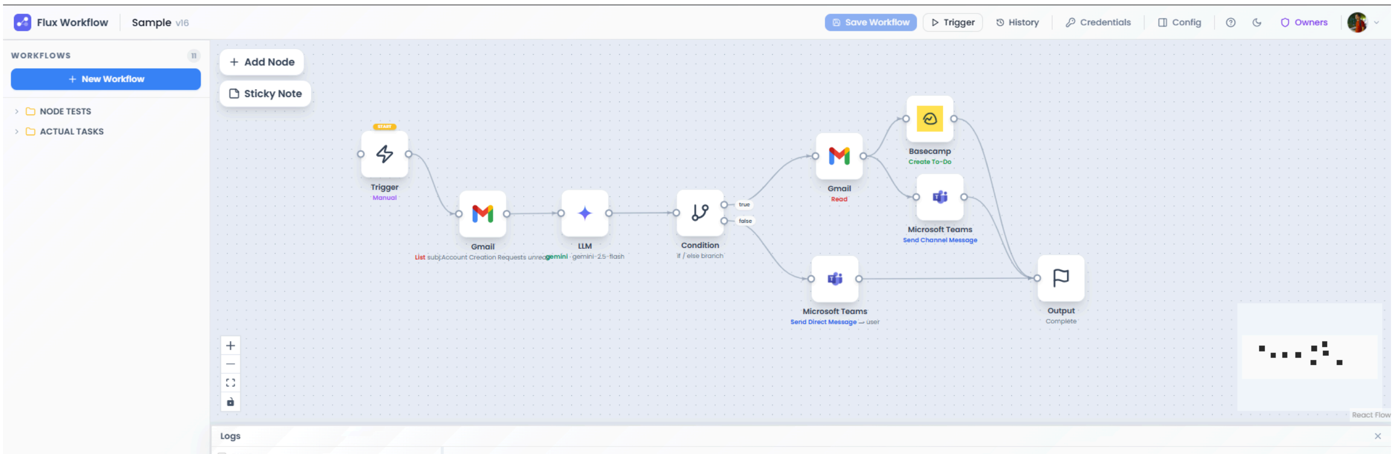
Technically yes, but with important controls.

**These four critical controls limit how this access is actually used:**

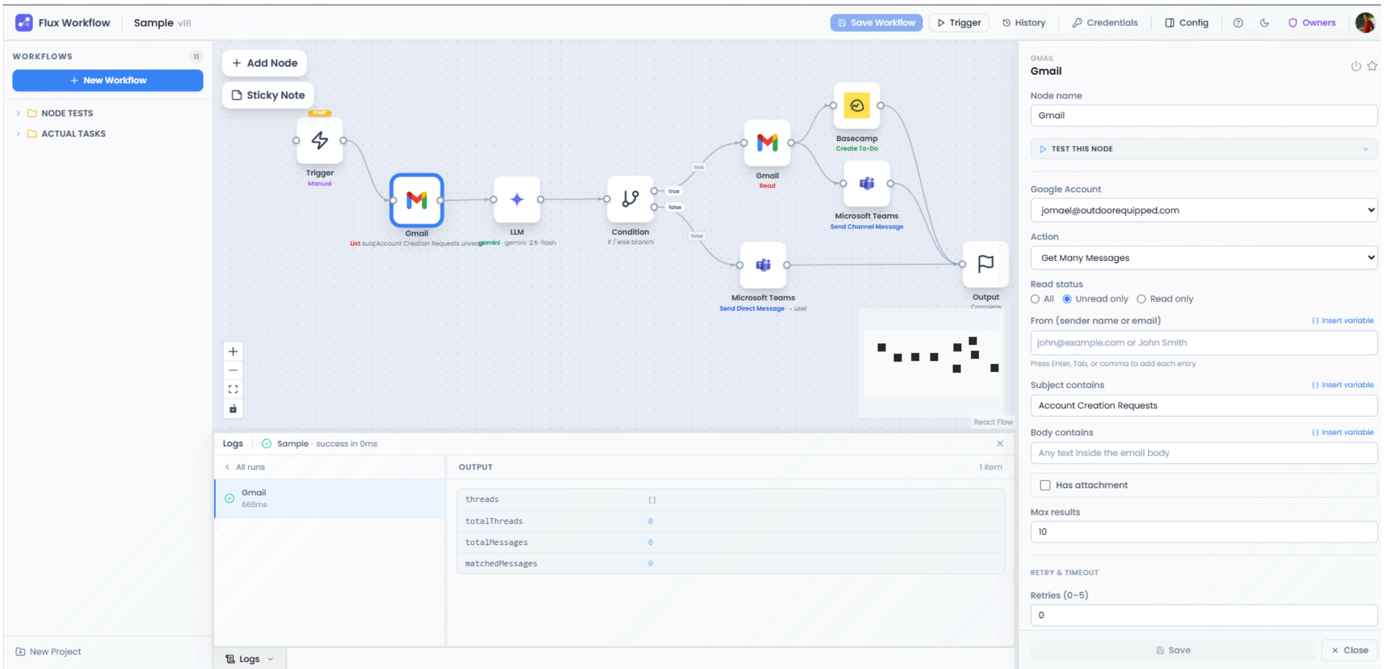
- Explicit user consent required: The access is granted only after the connecting user clicks through Google's standard OAuth consent screen. Google clearly shows what is being authorized before any token is issued. No access is obtained silently.
- Scoped to the connected account only. The token is tied to a single Google account (e.g., jomael@outdoorequipped.com). The platform cannot access any Google account that has not explicitly authorized it.
- Workflows must explicitly invoke it. The platform does not proactively scan or read Drive. A workflow node must be specifically configured by an authorized user - to perform a Drive action (list, download, upload, etc.). Nothing happens unless a workflow step calls it.
- The AI does not independently access Drive. The LLM node (OpenAI, Anthropic and Gemini) only processes data that is explicitly piped into it by a preceding workflow step. The AI has no autonomous connection to Google Drive and cannot initiate its own requests.

# Flux Workflow Snapshots

1. A basic workflow that processes incoming emails, uses Gemini to generate concise summaries, and executes follow-up tasks based on the summarized content.




2. A dedicated interface for managing Gmail (or other applications) configurations.



# Conclusion

1. **How much time will be saved?**

   Based on the types of tasks being automated, we project a savings of approximately 8 hours per person per week, which we will measure and validate post-deployment:

| Category | Time Saved per Week |
|---|---|
| Email triage and routing | ~2.5 hours |
| Cross-platform notifications | ~1 hour |
| Data report aggregation | ~2 hours |
| API lookups and manual actions | ~1.5 hours |
| AI-assisted summarization | ~1 hour |
| **Total per person** | **~8 hours/week** |

Across even a small team of 3–5 people, this translates to 24 - 40 hours of recovered capacity per week. The equivalent of adding a full-time contributor without additional headcount.

Beyond time savings, the platform delivers:

- Automated workflows run identically every time, eliminating human error
- Every execution is logged, providing a full paper trail
- New workflows can be added without writing custom scripts or engaging engineering
- Fully self-hosted, no per-task or per-seat SaaS fees
- Credentials and data never leave internal infrastructure

*"Flux Workflow converts hours of fragmented manual labor into reliable, observable, and repeatable automated processes — enabling the team to focus on higher-value work."*